Network Administration / System Administration

Homework 2

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0. VM Setup

Reference:

- unable to retrieve community.db: 404 errors (Arch Linux Forums)
- sudo pacman -Syu fails with error: failed to commit transaction (Stack Exchange)
- OpenSSH (ArchWiki)
- Enable SSH on Arch Linux

為了方便連線，我打算在 VM 當中配置 ssh server，使我可以利用 ssh 的方式進入虛擬機。步驟如下：

1. 執行 `run_vm.sh`，以 vnc 登入 VM 當中的 root 帳號。
2. 更新 archlinux 失敗，經查閱資料後，發現需要將 `/etc/pacman.conf` 中的 `[[community]]` 註解掉。
3. 更新 archlinux keyring

```
pacman -Sy archlinux-keyring
```

4. 更新 archlinux 並安裝 openssh：

```
pacman -Syu
pacman -S openssh
```

5. 啟動 sshd

```
systemctl enable sshd
systemctl start sshd
```

6. 離開 VM，並在 host 端修改 `run_vm.sh`，加入最後一行，使 host 端的 4321 port 轉發到 VM 的 22 port。

```
#!/bin/bash

echo "Starting VM at port $1"
echo "parameter is $(($1 - 5900))"

qemu-system-x86_64 -m 2G --enable-kvm \
        -vnc :"$(($1 - 5900))",password=on -monitor stdio \
        -drive file=disk1.qcow2,media=disk,index=0 \
        -drive file=disk2.qcow2,if=virtio,media=disk,index=1 \
        -drive file=disk3.qcow2,if=virtio,media=disk,index=2 \
        -drive file=disk4.qcow2,if=virtio,media=disk,index=3 \
        -drive file=disk5.qcow2,if=virtio,media=disk,index=4 \
        -drive file=disk6.qcow2,if=virtio,media=disk,index=5 \
        -drive file=disk7.qcow2,if=virtio,media=disk,index=6 \
        -drive file=disk8.qcow2,if=virtio,media=disk,index=7 \
        -drive file=disk9.qcow2,if=virtio,media=disk,index=8 \
        -drive file=disk10.qcow2,if=virtio,media=disk,index=9 \
        -net user,hostfwd=tcp::4321-:22 -net nic
```

7. 成功從工作站當中以 balu 的帳號 ssh 進去 VM。

```
ssh balu@localhost -p 4321
```

1. 那傢伙竟然敢無視窗

Reference:

- NTFS3
- Journaling file system (Wikipedia)
- How to mount NTFS in Linux without fuseblk (Stack Overflow)
- 使用 ntfs3 驱动替换 ntfs-3g 挂载 windows NTFS 分区
- 对 NTFS 硬盘进行读写操作；开机自动挂载硬盘 (CSDN)

NTFS 檔案系統可以滿足題目所要求，在雙系統中支援原生讀寫，並且在 Windows 11 下支援日誌功能。為了格式化 NTFS 檔案系統，需要先安裝 `ntfs-3g` 套件。我完成此題的步驟如下：

1. 在 VM 當中安裝 `ntfs-3g` 套件

```
sudo pacman -S ntfs-3g
```

2. 創立掛載點

```
sudo mkdir /mnt/myusb
```

3. 格式化硬碟

```
sudo mkfs.ntfs /dev/vdi2
```

4. 取得 UUID

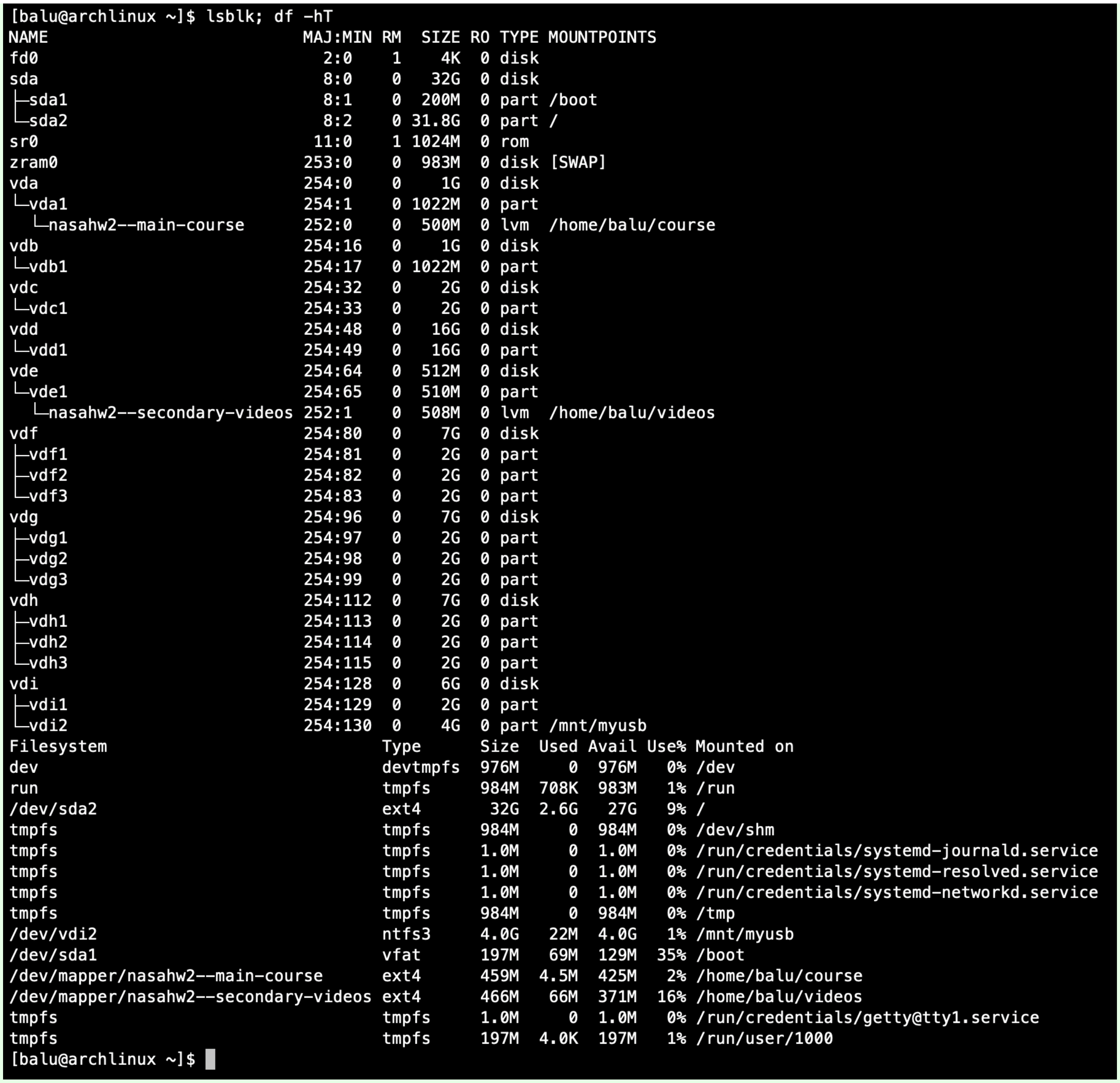
```
sudo blkid /dev/vdi2
```

Figure 1: Screenshot of `sudo blkid /dev/vdi2`

5. 配置自動掛載硬碟: 修改 /etc/fstab
6. 重開機，即可看到硬碟已經掛載

Ans.

Figure 2: Screenshot of `cat /etc/fstab`



```
[balu@archlinux ~]$ lsblk; df -hT
NAME                           MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
fd0                              2:0    1    4K  0 disk
sda                              8:0    0   32G  0 disk
├─sda1                           8:1    0  200M  0 part /boot
└─sda2                           8:2    0 31.8G  0 part /
sr0                             11:0    1 1024M  0 rom
zram0                          253:0    0  983M  0 disk [SWAP]
vda                            254:0    0    1G  0 disk
└─vda1                         254:1    0 1022M  0 part
  └─nasahw2--main-course       252:0    0  500M  0 lvm  /home/balu/course
vdb                            254:16   0    1G  0 disk
└─vdb1                         254:17   0 1022M  0 part
vdc                            254:32   0    2G  0 disk
└─vdc1                         254:33   0    2G  0 part
vdd                            254:48   0   16G  0 disk
└─vdd1                         254:49   0   16G  0 part
vde                            254:64   0  512M  0 disk
└─vde1                         254:65   0  510M  0 part
  └─nasahw2--secondary-videos  252:1    0  508M  0 lvm  /home/balu/videos
vdf                            254:80   0    7G  0 disk
├─vdf1                         254:81   0    2G  0 part
├─vdf2                         254:82   0    2G  0 part
└─vdf3                         254:83   0    2G  0 part
vdg                            254:96   0    7G  0 disk
├─vdg1                         254:97   0    2G  0 part
├─vdg2                         254:98   0    2G  0 part
└─vdg3                         254:99   0    2G  0 part
vdh                            254:112  0    7G  0 disk
├─vdh1                         254:113  0    2G  0 part
├─vdh2                         254:114  0    2G  0 part
└─vdh3                         254:115  0    2G  0 part
vdi                            254:128  0    6G  0 disk
├─vdi1                         254:129  0    2G  0 part
└─vdi2                         254:130  0    4G  0 part /mnt/myusb
Filesystem                             Type      Size  Used Avail Use% Mounted on
dev                                    devtmpfs  976M     0  976M   0% /dev
run                                    tmpfs     984M  708K  983M   1% /run
/dev/sda2                              ext4       32G  2.6G   27G   9% /
tmpfs                                  tmpfs     984M     0  984M   0% /dev/shm
tmpfs                                  tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs                                  tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
tmpfs                                  tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-networkd.service
tmpfs                                  tmpfs     984M     0  984M   0% /tmp
/dev/vdi2                              ntfs3     4.0G   22M  4.0G   1% /mnt/myusb
/dev/sda1                              vfat      197M   69M  129M  35% /boot
/dev/mapper/nasahw2--main-course       ext4      459M  4.5M  425M   2% /home/balu/course
/dev/mapper/nasahw2--secondary-videos  ext4      466M   66M  371M  16% /home/balu/videos
tmpfs                                  tmpfs     1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs                                  tmpfs     197M  4.0K  197M   1% /run/user/1000
[balu@archlinux ~]$
```

Figure 3: Screenshot of `lsblk; df -hT`

□

2. 因為要換到新的 SWAP

Reference:

- Create a Linux Swap File (Linuxize)

新增 SWAP 檔案的步驟如下：

1. 建立 /newswap 檔案

```
sudo fallocate -l 4G /newswap
```

2. 設定檔案權限

```
sudo chmod 600 /newswap
```

3. 格式化檔案

```
sudo mkswap /newswap
```

4. 啟用 SWAP 檔案

```
sudo swapon /newswap
```

Ans.

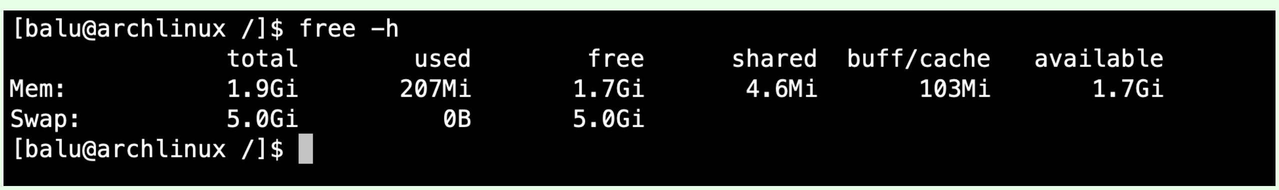


Figure 4: Screenshot of `free -h`

□

3. 為資料創造新的棲身之處

Reference:

- lvextend 命令 (LinuxCool)

使用指令如下：

```
sudo lvextend -L 1GiB /dev/nasahw2-main/course
```

Ans.

```
[root@archlinux /]# lsblk; df -hT
NAME                          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
fd0                             2:0    1    4K  0 disk
sda                             8:0    0   32G  0 disk
├─sda1                          8:1    0  200M  0 part /boot
└─sda2                          8:2    0 31.8G  0 part /
sr0                            11:0    1 1024M  0 rom
zram0                         253:0    0  983M  0 disk [SWAP]
vda                           254:0    0    1G  0 disk
└─vda1                        254:1    0 1022M  0 part
  └─nasahw2--main-course      252:0    0    1G  0 lvm  /home/balu/course
vdb                           254:16   0    1G  0 disk
└─vdb1                        254:17   0 1022M  0 part
  └─nasahw2--main-course      252:0    0    1G  0 lvm  /home/balu/course
vdc                           254:32   0    2G  0 disk
└─vdc1                        254:33   0    2G  0 part
vdd                           254:48   0   16G  0 disk
└─vdd1                        254:49   0   16G  0 part
vde                           254:64   0  512M  0 disk
└─vde1                        254:65   0  510M  0 part
  └─nasahw2--secondary-videos 252:1    0  508M  0 lvm  /home/balu/videos
vdf                           254:80   0    7G  0 disk
├─vdf1                        254:81   0    2G  0 part
├─vdf2                        254:82   0    2G  0 part
└─vdf3                        254:83   0    2G  0 part
vdg                           254:96   0    7G  0 disk
├─vdg1                        254:97   0    2G  0 part
├─vdg2                        254:98   0    2G  0 part
└─vdg3                        254:99   0    2G  0 part
vdh                           254:112  0    7G  0 disk
├─vdh1                        254:113  0    2G  0 part
├─vdh2                        254:114  0    2G  0 part
└─vdh3                        254:115  0    2G  0 part
vdi                           254:128  0    6G  0 disk
├─vdi1                        254:129  0    2G  0 part
└─vdi2                        254:130  0    4G  0 part /mnt/myusb
Filesystem                            Type      Size  Used Avail Use% Mounted on
dev                                   devtmpfs  976M     0  976M   0% /dev
run                                   tmpfs     984M  712K  983M   1% /run
/dev/sda2                             ext4       32G  6.6G   23G  23% /
tmpfs                                 tmpfs     984M     0  984M   0% /dev/shm
tmpfs                                 tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs                                 tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
tmpfs                                 tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-networkd.service
tmpfs                                 tmpfs     984M     0  984M   0% /tmp
/dev/vdi2                             ntfs3     4.0G   22M  4.0G   1% /mnt/myusb
/dev/sda1                             vfat      197M   69M  129M  35% /boot
/dev/mapper/nasahw2--main-course      ext4      459M  4.5M  425M   2% /home/balu/course
/dev/mapper/nasahw2--secondary-videos ext4      466M   66M  371M  16% /home/balu/videos
tmpfs                                 tmpfs     1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs                                 tmpfs     197M  4.0K  197M   1% /run/user/1000
[root@archlinux /]#
```

Figure 5: Screenshot of `lsblk; df -hT`

□

4. 我有拜託妳別把我的作業告訴其他人了吧

Reference:

- How to Encrypt a Drive in Linux (Medium)
- How to add a passphrase, key, or keyfile to an existing LUKS device (Red Hat)
- Linux 使用 LUKS 加密 Ext4 檔案系統

我做的步驟如下：

1. 以 root 進入 VM。
2. 切出一塊新的 lvm，作為加密區域

```
lvcreate -L 800MiB -n homework nasahw2-main
```

3. 加密 `/dev/nasahw2-main/homework`

```
cryptsetup luksFormat /dev/nasahw2-main/homework
```

4. 開啟加密區域

```
cryptsetup open /dev/nasahw2-main/homework homework
```

5. 格式化加密區域

```
mkfs.ext4 /dev/mapper/homework
```

6. 掛載加密區域

```
mount /dev/mapper/homework /home/balu/homework/
```

7. 設定 keyfile

```
cryptsetup luksAddKey /dev/nasahw2-main/homework /home/balu/lvm_key
```

8. 設定開機自動解密，在 `/etc/crypttab` 中加入底下這行：

```
homework /dev/nasahw2-main/homework /home/balu/lvm_key
```

9. 設定開機自動 mount，在 `/etc/fstab` 中加入底下這行：

```
/dev/mapper/homework /home/balu/homework ext4 defaults 0 2
```

10. 重開機，確認加密區域已經掛載

Ans.

```
[root@archlinux balu]#  lsblk; df -hT
NAME                          MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
fd0                             2:0    1    4K  0 disk
sda                             8:0    0   32G  0 disk
├─sda1                          8:1    0  200M  0 part  /boot
└─sda2                          8:2    0 31.8G  0 part  /
sr0                            11:0    1 1024M  0 rom
zram0                         253:0    0  983M  0 disk  [SWAP]
vda                           254:0    0    1G  0 disk
└─vda1                        254:1    0 1022M  0 part
  └─nasahw2--main-course      252:0    0    1G  0 lvm   /home/balu/course
vdb                           254:16   0    1G  0 disk
└─vdb1                        254:17   0 1022M  0 part
  ├─nasahw2--main-course      252:0    0    1G  0 lvm   /home/balu/course
  └─nasahw2--main-homework    252:1    0  800M  0 lvm
    └─homework                252:3    0  784M  0 crypt /home/balu/homework
vdc                           254:32   0    2G  0 disk
└─vdc1                        254:33   0    2G  0 part
vdd                           254:48   0   16G  0 disk
└─vdd1                        254:49   0   16G  0 part
vde                           254:64   0  512M  0 disk
└─vde1                        254:65   0  510M  0 part
  └─nasahw2--secondary-videos 252:2    0  508M  0 lvm   /home/balu/videos
vdf                           254:80   0    7G  0 disk
├─vdf1                        254:81   0    2G  0 part
├─vdf2                        254:82   0    2G  0 part
└─vdf3                        254:83   0    2G  0 part
vdg                           254:96   0    7G  0 disk
├─vdg1                        254:97   0    2G  0 part
├─vdg2                        254:98   0    2G  0 part
└─vdg3                        254:99   0    2G  0 part
vdh                           254:112  0    7G  0 disk
├─vdh1                        254:113  0    2G  0 part
├─vdh2                        254:114  0    2G  0 part
└─vdh3                        254:115  0    2G  0 part
vdi                           254:128  0    6G  0 disk
├─vdi1                        254:129  0    2G  0 part
└─vdi2                        254:130  0    4G  0 part  /mnt/myusb
Filesystem                            Type      Size  Used Avail Use% Mounted on
dev                                   devtmpfs  976M     0  976M   0% /dev
run                                   tmpfs     984M  732K  983M   1% /run
/dev/sda2                             ext4       32G  6.6G   23G  23% /
tmpfs                                 tmpfs     984M     0  984M   0% /dev/shm
tmpfs                                 tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs                                 tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
tmpfs                                 tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-networkd.service
tmpfs                                 tmpfs     984M     0  984M   0% /tmp
/dev/vdi2                             ntfs3     4.0G   22M  4.0G   1% /mnt/myusb
/dev/sda1                             vfat      197M   69M  129M  35% /boot
tmpfs                                 tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-cryptsetup@homework.service
/dev/mapper/nasahw2--main-course      ext4      459M  4.5M  425M   2% /home/balu/course
/dev/mapper/nasahw2--secondary-videos ext4      466M   66M  371M  16% /home/balu/videos
/dev/mapper/homework                  ext4      755M  220K  700M   1% /home/balu/homework
tmpfs                                 tmpfs     1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs                                 tmpfs     197M  4.0K  197M   1% /run/user/1000
[root@archlinux balu]#
```

Figure 6: Screenshot of `lsblk; df -hT`

□

5. 快照真的好難喔

Reference:

- LVM 系統快照設定教學 (iThome)
- Linux 解压缩 tar.zst 类型文件 (CSDN)
- How to keep or drop LVM snapshot (Server Fault)

我以 root 進入 VM，並進行以下操作：

1. 將 `/dev/vdc1` 設為 pv，並且加到 `nasahw2-main` vg 中

```
pvcreate /dev/vdc1
vgextend nasahw2-main /dev/vdc1
```

2. 建立 snapshot 的 lv

```
lvcreate -L 1GiB -s -n backup /dev/nasahw2-main/course
```

3. 掛載 `/dev/nasahw2-main/backup` 到 `/mnt/backup`

```
mkdir -p /mnt/backup
mount /dev/nasahw2-main/backup /mnt/backup
```

4. `lsblk` 結果如 Figure 7 所示。

5. 進行備份到 tar:

```
tar -I zstd -cvf /home/balu/backup.tar.zst /mnt/backup
```

6. 卸載 /mnt/backup，並刪除 snapshot。

```
umount /mnt/backup
lvremove /dev/nasahw2-main/backup
```

7. 執行 `lsblk; df -hT`，結果如 Figure 8 所示。

```
[root@archlinux balu]# lsblk
NAME                          MAJ:MIN RM   SIZE RO TYPE  MOUNTPOINTS
fd0                             2:0    1     4K  0 disk
sda                             8:0    0    32G  0 disk
├─sda1                          8:1    0   200M  0 part  /boot
└─sda2                          8:2    0  31.8G  0 part  /
sr0                            11:0    1  1024M  0 rom
zram0                         253:0    0   983M  0 disk  [SWAP]
vda                           254:0    0     1G  0 disk
└─vda1                        254:1    0  1022M  0 part
  └─nasahw2--main-course-real 252:4    0     1G  0 lvm
    ├─nasahw2--main-course    252:0    0     1G  0 lvm   /home/balu/course
    └─nasahw2--main-backup    252:6    0     1G  0 lvm   /mnt/backup
vdb                           254:16   0     1G  0 disk
└─vdb1                        254:17   0  1022M  0 part
  ├─nasahw2--main-homework    252:1    0   800M  0 lvm
  │ └─homework                252:3    0   784M  0 crypt /home/balu/homework
  └─nasahw2--main-course-real 252:4    0     1G  0 lvm
    ├─nasahw2--main-course    252:0    0     1G  0 lvm   /home/balu/course
    └─nasahw2--main-backup    252:6    0     1G  0 lvm   /mnt/backup
vdc                           254:32   0     2G  0 disk
└─vdc1                        254:33   0     2G  0 part
  └─nasahw2--main-backup-cow  252:5    0     1G  0 lvm
    └─nasahw2--main-backup    252:6    0     1G  0 lvm   /mnt/backup
vdd                           254:48   0    16G  0 disk
└─vdd1                        254:49   0    16G  0 part
vde                           254:64   0   512M  0 disk
└─vde1                        254:65   0   510M  0 part
  └─nasahw2--secondary-videos 252:2    0   508M  0 lvm   /home/balu/videos
vdf                           254:80   0     7G  0 disk
├─vdf1                        254:81   0     2G  0 part
├─vdf2                        254:82   0     2G  0 part
└─vdf3                        254:83   0     2G  0 part
vdg                           254:96   0     7G  0 disk
├─vdg1                        254:97   0     2G  0 part
├─vdg2                        254:98   0     2G  0 part
└─vdg3                        254:99   0     2G  0 part
vdh                           254:112  0     7G  0 disk
├─vdh1                        254:113  0     2G  0 part
├─vdh2                        254:114  0     2G  0 part
└─vdh3                        254:115  0     2G  0 part
vdi                           254:128  0     6G  0 disk
├─vdi1                        254:129  0     2G  0 part
└─vdi2                        254:130  0     4G  0 part  /mnt/myusb
[root@archlinux balu]#
```

Figure 7: Screenshot of `lsblk` after creating snapshot

```
[root@archlinux balu]# lsblk; df -hT
NAME                           MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
fd0                              2:0    1    4K  0 disk
sda                              8:0    0   32G  0 disk
├─sda1                           8:1    0  200M  0 part  /boot
└─sda2                           8:2    0 31.8G  0 part  /
sr0                             11:0    1 1024M  0 rom
zram0                          253:0    0  983M  0 disk  [SWAP]
vda                            254:0    0    1G  0 disk
└─vda1                         254:1    0 1022M  0 part
  └─nasahw2--main-course       252:0    0    1G  0 lvm   /home/balu/course
vdb                            254:16   0    1G  0 disk
└─vdb1                         254:17   0 1022M  0 part
  ├─nasahw2--main-course       252:0    0    1G  0 lvm   /home/balu/course
  └─nasahw2--main-homework     252:1    0  800M  0 lvm
    └─homework                 252:3    0  784M  0 crypt /home/balu/homework
vdc                            254:32   0    2G  0 disk
└─vdc1                         254:33   0    2G  0 part
vdd                            254:48   0   16G  0 disk
└─vdd1                         254:49   0   16G  0 part
vde                            254:64   0  512M  0 disk
└─vde1                         254:65   0  510M  0 part
  └─nasahw2--secondary-videos 252:2    0  508M  0 lvm   /home/balu/videos
vdf                            254:80   0    7G  0 disk
├─vdf1                         254:81   0    2G  0 part
├─vdf2                         254:82   0    2G  0 part
└─vdf3                         254:83   0    2G  0 part
vdg                            254:96   0    7G  0 disk
├─vdg1                         254:97   0    2G  0 part
├─vdg2                         254:98   0    2G  0 part
└─vdg3                         254:99   0    2G  0 part
vdh                            254:112  0    7G  0 disk
├─vdh1                         254:113  0    2G  0 part
├─vdh2                         254:114  0    2G  0 part
└─vdh3                         254:115  0    2G  0 part
vdi                            254:128  0    6G  0 disk
├─vdi1                         254:129  0    2G  0 part
└─vdi2                         254:130  0    4G  0 part  /mnt/myusb
Filesystem                             Type      Size  Used Avail Use% Mounted on
dev                                    devtmpfs  976M     0  976M   0% /dev
run                                    tmpfs     984M  740K  983M   1% /run
/dev/sda2                              ext4       32G  6.6G   23G  23% /
tmpfs                                  tmpfs     984M     0  984M   0% /dev/shm
tmpfs                                  tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs                                  tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
tmpfs                                  tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-networkd.service
tmpfs                                  tmpfs     984M     0  984M   0% /tmp
/dev/vdi2                              ntfs3     4.0G   22M  4.0G   1% /mnt/myusb
/dev/sda1                              vfat      197M   69M  129M  35% /boot
tmpfs                                  tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-cryptsetup@homework.service
/dev/mapper/nasahw2--main-course       ext4      459M  4.5M  425M   2% /home/balu/course
/dev/mapper/nasahw2--secondary-videos ext4      466M   66M  371M  16% /home/balu/videos
/dev/mapper/homework                   ext4      755M  220K  700M   1% /home/balu/homework
tmpfs                                  tmpfs     1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs                                  tmpfs     197M  4.0K  197M   1% /run/user/1000
[root@archlinux balu]#
```

Figure 8: Screenshot of `lsblk; df -hT` after removing snapshot

6. 好老舊喔

7. 我看還是再來合一次吧

8. 等一下，妳還沒回答我